

Two regimes of bioinspired memory: an architecture-search perspective

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Abstract—A robot learning across its service life must decide what to keep in fixed-size memory. The episodic buffer in e-MDB, a cognitive architecture for lifelong robot learning, discards the oldest entry whatever its value. We present EMBER (Emergent Memory-Based Encoding and Reactivation), which treats such a memory as a six-axis search space: how a trace is read, written, weighted by surprise, decayed, evicted, and reinforced. Their combinations give 576 *genotypes*, all settings of one engine; we evaluate all, including the e-MDB buffer. Each is scored on how many important-marked experiences survive a stream overrunning the memory many times over. The memory never sees the marks, only a surprise signal. We report four results. First, which mechanism matters depends on $r = K_{proto}/C$, recurring situations divided by capacity. Below $r = 1$ the write axis explains 73.0 % of the variance in retention; above it, 1.6 %, and eviction takes over. The same r marks the switch at three capacities: the ratio governs, not capacity itself. Second, the best genotype retains 0.983 of the marked experiences against 0.050 for the e-MDB buffer, differing on two axes only: surprise-weighted encoding and weakest-trace eviction. Neither alone changes anything. Third, a purely spiking circuit fails the reconstruction test the others pass: it was built for discrete patterns, not continuous vectors. Fourth, two preconditions appear only on real data. Merging needs repeated encounters to look alike; CIFAR-100 embeddings do not (0.401 against 0.85). Surprise must come from reward, not perceptual novelty, raising retention from 0.0 % to 52.6 %.

Index Terms—bioinspired memory, neural architecture search, Sparse Distributed Memory, engram consolidation, episodic memory, lifelong learning

I. INTRODUCTION

A cognitive robot that learns over its lifetime faces a memory problem that goes well beyond mere storage. It must decide what is worth keeping, keep that information accessible in proportion to its relevance, and retrieve it reliably even from a partial or degraded cue. These three requirements map onto three well-studied properties of biological memory: selective encoding modulated by surprise and novelty, use-regulated trace consolidation and decay, and content-addressable retrieval that is robust to noise.

The episodic buffer of the e-MDB architecture [1] —a fixed-capacity FIFO queue— satisfies none of the three. It keeps whatever was perceived most recently, treating every experience alike regardless of its importance, novelty or prediction error. Retrieval is sequential, not by content. When the buffer fills, the oldest episode is simply discarded: it is, in effect, a database with no criterion.

The bioinspired-memory literature offers several principled remedies. Kanerva’s Sparse Distributed Memory (SDM) [2] provides content-addressable retrieval over a distributed superposition of memory locations, with graceful degradation

under noise. Hebbian engram consolidation [3], [4] offers a biologically plausible mechanism for reinforcing traces through repeated co-activation. Prediction-error gating [5], grounded in the neurobiological role of dopamine, regulates how strongly a new experience is encoded as a function of how surprising it is. And selective-forgetting policies based on trace strength or utility [6] give a criterion for what to discard when capacity is exceeded.

But these mechanisms are almost always evaluated in isolation, embedded in architectures purpose-built to show them off, and compared against baselines that do not share their design space. That makes it hard —sometimes impossible— to tell whether a reported gain comes from the mechanism itself or from an incidental advantage of the evaluation protocol.

This work reports on EMBER (Emergent Memory-Based Encoding and Reactivation), a research program on long-term episodic memory for cognitive robots, and it inverts the usual approach. Rather than proposing a new architecture and arguing that it beats a simple baseline, we define a discrete design space in which each bioinspired mechanism is an axis, and we traverse it exhaustively. The e-MDB FIFO is not treated as a baseline external to the comparison: it is one more point inside the same search space. This choice turns a potentially circular comparison into a falsifiable question —given the full space, where does the deployed incumbent land, and which combination of mechanisms defines the performance frontier?

We then evaluate the bioinspired architectures that emerge from that analysis with a two-phase benchmark separating the ability to reconstruct a degraded cue —a prerequisite of any associative memory— from retaining the right episodes under continuous capacity pressure. Finally, we instrument the law and the salience mechanism on two real domains, perceptual embeddings and a navigation environment under partial observability, exposing preconditions no synthetic design reveals, one of which we correct.

Three terms recur throughout. A *surprise signal* is a scalar in $[0, 1]$ attached to an experience when it is written: either *novelty*, which the memory computes as one minus the similarity to the closest thing it holds, or *prediction error*, from a model of what was expected. The *salience gate* turns that signal into an encoding weight. *Rare-event retention* is the fraction of importance-tagged episodes still recoverable after a stream that exceeds capacity by more than an order of magnitude; the tag is ground truth used only for scoring, never visible to the memory, which is what keeps the measurement from being circular and why eviction by age is worst placed under it.

The main contributions of this work are:

- A *threshold law*: the ratio $r = K_{proto}/C$ between recurring experience prototypes and memory capacity determines which family of mechanisms governs performance, with a regime transition that reproduces independently across three distinct capacities.
- A demonstration that in a design space of gated mechanisms, main effects mislead. Modulating encoding strength by novelty and prediction error—the most-cited mechanism in the bioinspired-memory literature—explains barely 1.4 % of the variance as a main effect, yet conditioned on the one eviction policy that reads the strength signal it improves rare-event retention by a factor of 14.41. The frontier architecture differs from the deployed incumbent on two axes only, and neither is worth installing without the other.
- A two-phase benchmark that admits architectures by reconstruction before exposing them to capacity pressure, revealing that a pure spiking circuit (LIF + STDP) does not transfer to the continuous-vector domain without an intermediate encoding layer.
- Two preconditions that appear only on real data. Consolidation requires that the merge threshold be reachable in the domain, which on CIFAR-100 it is not: there is no r , nominal or effective, at which a compression regime exists. And the salience mechanism requires surprise to be anchored in the variable that decides what matters—reward—rather than in the perceptual novelty of the state; unlike the first, this one we identify and correct.

Each of the three data sources used below removes a convenience the previous one enjoyed: the synthetic generator is the only setting in which K_{proto} is known rather than estimated, CIFAR-100 removes the near-orthogonality of synthetic vectors, and MiniGrid removes both the hand-assembled stream and the hand-assigned surprise signal. The last two contributions are the preconditions that surfaced when a convenience was withdrawn.

II. RELATED WORK

A. Sparse Distributed Memory

Kanerva’s Sparse Distributed Memory [2] models associative memory on the high-dimensional structure of the nervous system. Storing information in one location is fragile; instead patterns are distributed across many overlapping *hard locations* with random addresses, so a noisy query still activates enough of the same ones to reconstruct the original. A write activates the locations within a similarity radius of the queried address; a read sums and thresholds those same locations.

Concretely, in the instantiation used here: 512 hard locations are drawn once as random unit vectors and never move. Writing an episode finds the 25 locations (5 %) most aligned with its key and adds $s \cdot k$ to each of their counters, where s is the encoding strength, so the episode is smeared across twenty-five locations and stored in none of them. Reading takes the query, sums the counters of its 25 nearest locations

and normalizes. A noisy query selects a slightly different active set, but an overlap of twenty of twenty-five still reconstructs the right pattern, so accuracy declines gradually rather than failing abruptly—the property that matters for a robot, whose sensors are never clean. Two adaptations should be stated. The address space is continuous and activation is by inner product rather than Hamming radius, which makes the radius self-adapting to local density; [7] extends the same intuition to hyperdimensional computing. And our implementation is a hybrid: it keeps both the distributed counters and an explicit list of traces, and a read returns the stored trace most similar to the reconstruction rather than the reconstruction itself. This preserves exact payloads, but it is a departure from the textbook model.

B. Engrams, Hebbian consolidation and STDP

The *engram*—the physical trace a memory leaves in neural tissue—was coined by Semon [8] and confirmed experimentally in modern neuroscience [3]. Its cellular mechanism is long-term potentiation: neurons that fire together repeatedly strengthen the synapse between them, the principle Hebb [4] distilled into his dictum. In EMBER’s ENN instantiation this is a matrix of fast Hebbian weights updated online without backpropagation: each write creates a trace or reinforces an existing one above a cosine threshold, with initial strength proportional to prediction error and exponential decay without reinforcement.

Spike-timing-dependent plasticity (STDP) [9] is the same rule at the level of the individual spike: a synapse strengthens when the presynaptic neuron fires just before the postsynaptic one, and weakens in the opposite order. This groups neurons into assemblies, leaving a structural trace that is precisely an engram. EMBER’s PoC-3 verified this on leaky integrate-and-fire neurons: after repeated presentation, intra-group weights rose well above inter-group weights, and a partial cue recruited the rest of its group without activating the others. Whether that mechanism extends to continuous input vectors is what Section IV tests.

C. Architecture search and prioritized replay

Neural architecture search [10] treats architectural decisions as optimizable variables. Its application to the memory mechanism itself is far less explored: Neural Turing Machines [11] and related differentiable-memory architectures [12] fix the memory type and search over read/write heads. We instead make the memory mechanism the search variable, over a fully enumerable space, so that no result can be attributed to an approximate optimizer.

The closest analogue outside neuroscience is deep reinforcement learning. Prioritized experience replay [13] weights which transition to revisit by temporal-difference error—the same “surprising things count more” principle that motivates prediction-error gating—and Neural Episodic Control [14] indexes a differentiable dictionary by similarity, analogous to SDM’s content-addressable read. In both, the priority and read mechanisms are chosen in advance. This work treats those

choices as the search space, and measures in which regime each one matters.

D. The e-MDB architecture and the LOLA problem

e-MDB (*epistemic Multilevel Darwinist Brain*) [1] is a ROS2-based cognitive architecture designed to address the LOLA problem (*Lifelong Open-ended Learning Autonomy*): building a robot that acquires and relates knowledge autonomously and continuously, with no predefined goals or task boundaries. It organizes its components as *Cognitive Nodes* with a common interface and uses an Episode object as the unit of experience. Its episodic-memory component, *EpisodicBuffer*, is a fixed-size deque: episodes are appended and the oldest discarded at the limit. There is no importance criterion, no similarity-based retrieval, and no consolidation. In the NAS formulation of Section III, this corresponds exactly to the genotype {*read*: nn, *write*: append, *strength*: constant, *decay*: 1.0, *evict*: fifo, *reinforce*: 0.0} —one point among 576 in the design space.

III. SEARCH OVER A BIOINSPIRED DESIGN SPACE

A. Design space and genotype encoding

A *genotype* in this work is a complete memory design: one choice on each of six axes, where every axis is a decision an engineer building an episodic memory has to make anyway, usually by intuition. How is a stored experience retrieved —by its single closest match, by a few, or by a whole neighborhood? Is a repeated experience a new memory or a reinforcement of an existing one? Does every experience encode with the same weight, or does surprise weight it? Do unreinforced memories fade? When memory fills, what is discarded? And does recalling a memory strengthen it? Table I lists the six axes, their options, and the biological mechanism each one stands for; Fig. 1 shows how they compose into a single write. Enumerating the Cartesian product —576 genotypes— means every combination is tested, including the ones no designer would think to try, and the incumbent is evaluated by the same procedure as everything else rather than as an external baseline.

The biological vocabulary has an exact computational referent. A *genotype* is a six-valued configuration vector, each *axis* is one parameter of it, and the *phenotype* is the memory obtained by instantiating a single parameterized engine with that vector. The 576 are therefore not independently written systems but configurations of one engine, in which every mechanism is implemented exactly once and shared by every genotype that selects it. This is what makes the comparison sound: the same code path runs for every point, so any score difference is attributable to the axis settings alone. An earlier iteration kept separate implementations for the search and the benchmark, whose eviction rules had silently diverged, so tables presented as comparable were not measuring the same quantity. Two conventions follow, both standard: *architecture search* names the method in the sense of the NAS literature [10], where candidates are instantiated from shared machinery,

TABLE I
BIOINSPIRED DESIGN SPACE. THE CARTESIAN PRODUCT YIELDS 576
CANDIDATE GENOTYPES, EXHAUSTIVELY ENUMERABLE.

Axis	Options	Biological root
<i>read</i>	nn / topk3 / radius	Episodic retrieval vs. activation circle (Kanerva)
<i>write</i>	append / merge	Trace creation vs. Hebbian merge consolidation
<i>strength</i>	constant / novelty / pred_error / both	LTP gating by salience and surprise (dopamine)
<i>decay</i>	1.0 / 0.995 / 0.98	Synaptic weakening without reinforcement
<i>evict</i>	fifo / min_strength / min_utility / random	Forgetting by age vs. by importance
<i>reinforce</i>	0 / 0.5	LTP by reactivation

and *genotype* names an individual candidate. We reserve *architecture* for Section IV, where it denotes four memory substrates differing in how a trace is physically represented.

The *read* axis operationalizes how past experiences are retrieved: *nn* returns the single closest stored episode; *topk3* aggregates the three closest; *radius* implements the full SDM read, superposing all traces within a similarity radius relative to the best match. The *write* axis controls trace creation: *append* always creates a new trace; *merge* first checks whether a sufficiently similar trace already exists (cosine similarity ≥ 0.85) and, if so, reinforces it instead of creating a redundant copy —the computational analogue of Hebbian consolidation. The *initial-strength* axis —the *strength gate*, which elsewhere in the text we also call the *salience gate* when the emphasis is on which biological signal opens it— modulates how much weight a new experience receives at encoding time: *constant* assigns a fixed weight; the other three options introduce a biological signal —novelty, prediction error, or the two combined. The *decay* axis determines whether the strength of stored traces erodes over time without reinforcement. The *eviction* axis controls what is forgotten when memory fills: *fifo* discards the oldest (the e-MDB default), *min_strength* discards the weakest trace, *min_utility* discards the least-queried, and *random* discards at random. The *read-reinforcement* axis models LTP by reactivation: each time an episode is retrieved, its strength increases.

An early version of this space had two silently dead axes: the pilot reported 576 genotypes but only 96 were functionally distinct, because *read* and *reinforce* changed the outcome for no pair of siblings. Both were repaired —*read* because its three modes now differ, *reinforce* because the tasks interleave reads between writes— and an axis-liveness audit runs in continuous integration so no axis can die again unnoticed.

B. Axis observability

An axis that changes the outcome nowhere is measuring nothing, and any conclusion about that mechanism’s irrelevance is really a conclusion about the instrument. Before reporting effects we verify all six axes are observable, by measuring the maximum performance difference between genotypes differing only in that axis.

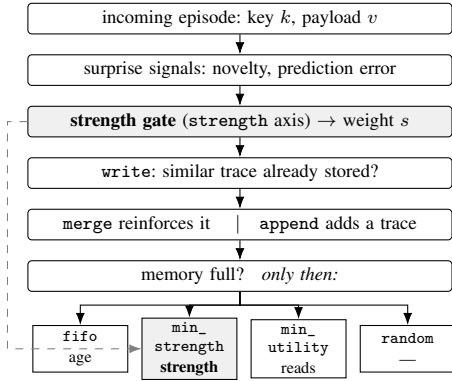


Fig. 1. The write path, and why the salience gate is conditional. The **strength** axis assigns a weight s to every incoming episode, but s is inert unless something downstream reads it. The dashed line marks the only consumer: of the four eviction policies, **min_strength** is the one that consults trace strength, so in three quarters of the design space the salience signal is computed and discarded. Eviction runs only when the memory is already full, which is why the regime transition of Section III falls where it does.

Write mode is the strongest main effect (56.6 %), followed by the eviction policy (17.0 %). Decay and initial strength explain much less as main effects (1.8 % and 1.4 % respectively), and read and read-reinforcement come out near zero (0.0 % and 0.0 %). The difference from the pilot is that both now have strictly positive observability (0.119 and 0.206): a null η^2 with positive observability does not say the mechanism does not matter, it says its effect is not a main effect —Section III-D shows exactly that case for initial strength.

The e-MDB FIFO —genotype {read: nn, write: append, strength: constant, decay: 1.0, evict: fifo, reinforce: 0.0}— scores 0.106, in the range 416–469 of 576, with 54 ties. It is not the absolute floor of the genotype space: below its pessimistic rank of 469 there are genotypes that score strictly worse, almost all combining criterion-free append writing with minimum-utility eviction or topk3 reads over compressed memory. The frontier of the space scores 0.818: a factor of almost eight over the incumbent.

C. The regime transition

The 56.6 % of the write axis is not primarily a property of the merge mechanism; it is a property of the relationship between the number of recurring experience prototypes and memory capacity. We define the prototype-to-capacity ratio $r = K_{proto}/C$ and sweep the genotype space over a grid of capacities and ratios, controlling visits per prototype so that the ratio is not confounded with stream recurrence.

The transition from “the write axis explains most of the variance” to “the eviction axis does” falls abruptly near $r = 1$ across the three evaluated capacities (10, 20 and 40), independently of K_{proto} : the estimated crossover (geometric mean over the transition cell) is 0.88, with interval [0.87, 0.89]. The mechanism is simpler than the statistics that localize it. Eviction never runs while the memory has room: a trace is discarded only when a write would exceed capacity. Below

TABLE II
NEITHER MECHANISM WORKS ALONE. MEANS OVER THE THREE BATTERY TASKS.

Genotype	mean	T1	T2	T3
e-MDB incumbent	0.106	0.050	0.269	0.000
+ salience gate only	0.106	0.050	0.269	0.000
+ min_strength only	0.106	0.050	0.269	0.000
+ both (frontier)	0.818	0.983	0.469	1.000

$r = 1$ the recurring experience *fits*, merging compresses it into fewer slots, and the freed slots become available for the rare, so what matters is how one writes. Above $r = 1$ there is nothing to compress, memory is saturated, eviction runs on nearly every write, and the question reduces to which policy best preserves what matters. We call these the *compression* and *selection* regimes, and the claim of this section is that which of the two a memory is operating in, rather than any property of the mechanisms themselves, determines which mechanism matters.

Two cautions on reading Fig. 2. First, η^2 measures attribution, not activity: it reports how much of the spread in retention across the 576 genotypes is explained by which option each takes on that axis, not how often the operation runs — every genotype in a cell receives the identical stream and performs the identical number of writes. Second, the two curves are not the whole decomposition: above the threshold they jointly explain roughly a fifth of the variance, the remainder belonging to the strength gate, decay and their interaction with eviction —the structure Section III-D isolates. That is what the sweep records, not noise: within a seed the evaluation is deterministic.

D. The interaction between salience and eviction

The 1.4 % main effect of initial strength is misleading read on its own. Strength modulation influences behavior only when a mechanism *reads* that signal, and of the four eviction policies only **min_strength** does. Conditioning on {evict: min_strength, write: append}, a constant gate leaves the policy nothing to rank by (3.6 %), whereas novelty and prediction error together make it selective (52.0 %): a conditional improvement of $14.41\times$, entirely attributable to this interaction.

The same interaction can be shown without statistics, on four concrete genotypes. The frontier differs from the e-MDB incumbent on exactly two of the six axes: the strength gate (constant $\rightarrow$ both) and the eviction policy (fifo $\rightarrow$ min_strength). Changing either alone leaves the architecture *bit-identical* to the incumbent, in every combination of the remaining axes (Table II). The reason is exact rather than statistical: under a constant gate with no read reinforcement every trace carries strength 1.0, so selecting the minimum always returns the oldest surviving trace — **min_strength** does not approximate FIFO, it *is* FIFO. Symmetrically, a salience gate feeding a policy that never reads strength computes a weight nothing consumes.

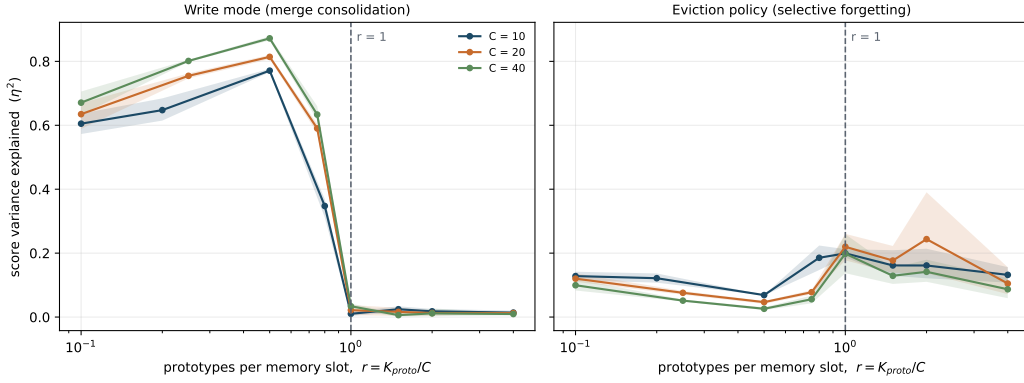


Fig. 2. Where the regime transition falls. Share of the variance in rare-event retention, across the 576 genotypes, attributable to the write axis and to the eviction axis, as a function of $r = K_{proto}/C$. This is a falsification test rather than a performance plot: if absolute capacity governed the transition the three curves would cross at three different points, and if stream redundancy governed it, holding visits per prototype fixed would have flattened the effect. Neither happens. Note that η^2 is direction-blind — it identifies which axis moves the outcome, not which setting is preferable: the frontier architecture uses *append*, not *merge*. Bands: bootstrap confidence interval over seeds.

E. The frontier architecture

The best-scoring point of the space is {read: nn, write: append, strength: both, decay: 1.0, evict: min_strength, reinforce: 0.0}, at 0.818 against 0.106. In practice: every experience is stored as its own trace, written with a weight proportional to how surprising it was. Weights never decay and are never revised on recall, so importance is fixed at encoding. Recall returns the closest trace, and when memory fills the lowest-weight trace is discarded — by construction the least surprising thing held.

Two properties matter more than the rank. It *appends*: the axis that dominates the variance below $r = 1$ is the write mode, but the winning setting never consolidates. Merging compresses routine into fewer slots, yet accumulates strength geometrically on whatever recurs, so a prototype visited thirty times outweighs a rare event encoded once under maximum surprise. And it is *regime-neutral*: never depending on merging, it does not care whether r is above or below the threshold, and always consulting the importance signal it behaves identically in both. That is what makes it a deployable recommendation rather than the winner of one cell — a robot running this genotype need not know, or estimate, its own r .

IV. COMPARATIVE ARCHITECTURE EVALUATION

Architecture search identifies the best-performing combination of axes, but it does not evaluate the architectures as standalone memory systems. In particular, it does not test whether each one can perform the fundamental operation any associative memory must support: reconstructing a complete pattern from a noisy or partial cue. This section describes a two-phase benchmark that first admits architectures by reconstruction fidelity (R1–R4, no capacity pressure) and then evaluates them under continuous capacity pressure (T1–T3), over the same four architectures —SDM, ENN (pure Hebbian memory), Spiking-SDM (SDM with LIF-neuron activation), and a pure spiking circuit (LIF + STDP)— plus the e-MDB

FIFO as a baseline sharing the same design space. Table III summarizes both phases.

Every architecture except the pure spiking circuit clears the admission threshold of 0.50 (0.459 against a gate of 0.50). ENN and FIFO get exactly the same gate score: with no capacity pressure, ENN’s merge threshold (0.85) is rarely exceeded between near-orthogonal vectors, so every write creates a new trace and both architectures reduce to nearest-neighbor search over the full set of stored keys. The difference between them emerges only under capacity pressure with structured streams, in phase 2.

The pure spiking circuit fails pattern reconstruction and noise robustness despite forming distinguishable assemblies (perfect $A \rightarrow B$ interference). The cause is a domain mismatch: the circuit was designed and validated on discrete neural activation patterns; over a continuous `float32` vector, inner-product neuron selection is sensitive to small perturbations, and a noisy version of a stored vector activates a partially different set of neurons. This result is not a flaw of the spiking mechanism itself —it is the direct motivation for EMBER’s PoC-5: an integrated core combining SDM (for the continuous-to-distributed mapping) with ENN (for the trace lifecycle).

One property of the phase-2 protocol should be stated before its numbers are read. The stream carries twenty rare items into a memory of capacity twenty, and the generator draws the prediction error of common and rare experiences from non-overlapping ranges. An architecture that gates on prediction error and evicts by minimum strength therefore *should* end holding exactly the rare set: the retention component of a perfect score is close to arithmetically forced, and confirms the mechanism is wired correctly rather than that the problem was hard. What the score does test independently is retrieval — each trace must still be recovered from the distributed superposition at similarity ≥ 0.85 . The informative comparison is therefore not SDM against the FIFO but SDM against ENN, which uses the identical gate and eviction policy and retains

TABLE III

TWO-PHASE EVALUATION. R1–R4 MEASURE RECONSTRUCTION WITH NO CAPACITY PRESSURE AND DECIDE ADMISSION; T1–T3 MEASURE RETENTION UNDER PRESSURE.

Arch.	R1	R2	R3	R4	Mean	Gate	T1	T2	T3
SDM	0.672	0.788	1.000	0.978	0.859	✓	1.000	0.365	0.875
ENN	0.691	0.938	1.000	1.000	0.907	✓	0.750	0.344	0.375
Spiking	0.190	0.146	1.000	0.499	0.459	×	—	—	—
SpikingSDM	0.507	0.433	0.950	0.588	0.619	✓	0.675	0.158	0.719
FIFO	0.691	0.938	1.000	1.000	0.907	✓	0.038	0.254	0.000

less. The threshold grid of Section III, with rare count at half the capacity, does not share this ceiling.

In phase 2, SDM achieves perfect rare-event retention (1.000, $\sigma = 0.000$ across the four seeds): rare events receive high initial strength from prediction error, and minimum-strength eviction always displaces the routine first. ENN, with the same strength gate and the same eviction policy, retains less (0.750, also with no seed-to-seed spread): its merge mechanism, reinforced repeatedly by routine visits, can accumulate more strength than a rare event written only once, though this effect is less severe than in earlier designs because the intra-prototype spread of the current streams triggers merging less often. Spiking-SDM retains 0.675, and the FIFO —with no importance criterion at all— is the floor among architectures (0.038, $\sigma = 0.041$: the floor is consistent across seeds, not an artifact of a single run, though noisier than the rest because of how close it is to zero). With a block-differentiated prediction-error signal, the sequential-interference task gives 0.875 for SDM, 0.375 for ENN, and 0.000 for the FIFO —all three with no seed-to-seed spread. This is the comparison that separates the three architectures by their ability to resist catastrophic forgetting.

V. VALIDATION ON REAL PERCEPTUAL EMBEDDINGS

The preceding experiments run on random Gaussian vectors in $\mathbb{R}^{32}$, nearly orthogonal at that dimension. This reduces cross-interference and could overestimate retrieval accuracy relative to real embeddings, which have correlated structure and uneven cluster density. We therefore instrument the same grid on streams built from CIFAR-100 [15] embeddings extracted with a pretrained ResNet-18 [16] and reduced by PCA to the working dimension.

The result is not that the threshold shifts: it is that, with this extractor, the write axis is inoperative in all five cells of the sweep, explaining between 0.0 % and 0.4 % of the variance regardless of the ratio. The deciding quantity is *intra-prototype similarity*: how alike two encounters with the same kind of thing look to the memory, as a cosine. Consolidation is possible only when it clears the merge threshold, since merging is triggered by exactly that comparison — if two photographs of one class do not look alike enough, the memory cannot recognize them as the same situation and stores both. The measured median is 0.401, well below the merge threshold (0.85), so merging almost never triggers and $K_{effective}$ —the traces the routine occupies after consolidation— grows without bound instead of converging to K_{proto} : from 84 to 1435 traces

as nominal K rises from 5 to 80, with capacity fixed at 20. The synthetic domain, in the same run, does cross normally ($r \approx 0.77$), confirming that the engine and the rest of the grid behave as expected.

Intra-prototype similarity is not a term in r and does not move the threshold; what it decides is whether r is well defined at all. Where consolidation is reachable but partial, it acts on the ratio indirectly through $K_{effective}$: visits below the merge threshold fragment a prototype into several traces, so the effective ratio exceeds the nominal one and the law holds when stated over it. Where consolidation is unreachable there is no stable $K_{effective}$ — it grows with stream length instead of converging, so an effective ratio would describe how long the robot has run, not its environment.

This precondition is therefore not a special case of rewriting the law over an effective ratio: it is a separate binary check that comes first, and on CIFAR-100 it already fails, so no recomputation of r —nominal or effective— produces a compression regime where there is none.

On a real stream K_{proto} must be estimated. We cluster the keys over a sweep of candidate counts and take the count maximizing the mean *silhouette coefficient*, which for each vector compares its distance to its own cluster against its distance to the nearest other one: near 1 means tight, well-separated clusters, near 0 means the partition is arbitrary. Clustering returns groups whether or not any exist —on isotropic noise the whole sweep scores 0.04–0.07 with no peak, yet still yields an argmax— so we require a silhouette of at least 0.15 before treating r as defined, and report a bootstrap interval rather than a point integer.

On CIFAR-100 that estimation becomes uninformative: the intervals reach the ceiling of the search in several cells, consistent with the absence of compact cluster structure in this domain. This is a further reason to treat merge reachability —measured directly as a similarity comparison, with no clustering involved— as the robust diagnostic, rather than a derived $\hat{K}$.

VI. VALIDATION IN A REAL ENVIRONMENT UNDER PARTIAL OBSERVABILITY

CIFAR-100 validates the threshold law on real data, but it is still a hand-built stream from static embeddings. A robot does not receive a stream: it receives the consequences of its own actions, with partial observation and temporal structure that no generator reproduces. We instrument the `ember.envs` adapter on MiniGrid-MemoryS13-v0 [17], a navigation environment

designed specifically to demand memory: the agent must remember an object seen at the start of the episode in order to choose correctly at the end. We compare the frontier genotype from Section III against the FIFO with `t1_rare_retention`—the same task that measures rarity throughout the rest of the program— across three conditions that change one variable at a time (Table IV).

Baseline. With a uniform random policy, both the frontier and the FIFO sit at the rare-event retention floor. The cause is not a retrieval error—with no capacity pressure the frontier recovers any stored trace with similarity 1.0—but that the one-step prediction error, computed by an online linear transition model over the observation, does not separate the rewarded from the merely novel: 75.4 % of common experiences have a prediction error equal to or greater than the least surprising rewarded event, measured over 38 rewarded events. This quantity is the *overlap* reported in Table IV: the smaller it is, the more cleanly the surprise signal separates what mattered from what did not.

Ruling out data scarcity. The simplest explanation must be ruled out first. A policy biased toward moving forward over turning—which does not touch the surprise computation, only what gets visited— raises the rewarded-episode count from 38 to 156 on the same step budget. Retention stays at the floor, and the overlap does not improve (73.5 %): quadrupling the data without touching the signal changes nothing, so the problem is not a small sample.

The signal, not the quantity. The strength gate is motivated by the dopaminergic account of prediction error [5], but that account is about *reward*-prediction error, not perceptual. We replace the transition model with a reward predictor—same online linear architecture, same random policy as the baseline, changing only what is predicted—and the overlap falls to 5.6 %. The frontier’s retention rises from 0.0 % to 52.6 %—with 38 rare events per condition, a 95 % Wilson interval of [37.3 %, 67.5 %]— while the FIFO—which does not read the strength signal at all— stays at 2.6 % ([0.5 %, 13.5 %]), just as in the baseline. The two intervals do not overlap despite the small sample. That only the architecture consulting the strength signal improves rules out an artifact of the predictor change: it is the salience gate working, now given a signal correlated with what matters.

The salience mechanism does transfer to a real environment under partial observability—not automatically, but under an identifiable and correctable precondition: surprise must be anchored in the variable that decides what matters, which in an environment with reward is reward itself.

VII. DISCUSSION

A. The threshold law and its practical implication

The ratio $r = K_{proto}/C$ has a direct operational reading. In an early, structured environment—a warehouse, a lab—the robot sees the same objects and corridors repeatedly and $r < 1$, where consolidation is efficient. In a heterogeneous environment r grows above 1 and selective eviction by relevance

becomes the dominant requirement. The design recommendation is not to pick a regime: the frontier architecture works in both, because it never depends on merging—which fails above $r = 1$ — and always exploits the importance signal, which is regime-neutral by construction.

B. The spiking circuit and the motivation for PoC-5

That the spiking circuit does not pass the reconstruction gate is a result of the design space, not a deficiency of the mechanism: PoC-3 established that LIF+STDP circuits form reliable engrams and complete patterns in their native domain. The gap is that the bridge between continuous vectors and discrete assemblies does not yet exist, which is the goal of EMBER’s PoC-5: SDM for the continuous-to-distributed mapping, ENN for the trace lifecycle, as a single *Cognitive Node*.

C. The FIFO contrast, and the precondition as a diagnostic

Mapping the e-MDB FIFO to a genotype is deliberately generous: the real `EpisodicBuffer` has no content-addressable retrieval at all. Even with that advantage it lands in the bottom quartile of the genotype space, and among admitted architectures it is the floor at retaining rare events—though not at reconstruction, where it scores as well as any. This is not a criticism of e-MDB, designed with a different primary goal; it is a precise claim that the episodic buffer as implemented is inadequate for the LOLA operating regime, with a quantified gap to close.

The finding of Section V gives the robot something more useful than a generalization warning: a check it can run online. Intra-prototype similarity is measured directly on the vectors the memory already receives, and comparing it against the merge threshold decides—before the write axis matters at all— whether consolidation is worth attempting. A robot monitoring this can tell, with no external instrumentation, whether merging helps or is wasted work.

D. The source of surprise matters as much as its existence

Section VI generalizes this beyond merging. The synthetic and CIFAR-100 designs share a silent assumption: that the surprise signal arrives aligned with what the robot needs to remember. That is free when a generator sets it by hand; in a real environment one must choose *what to predict*, and that choice decides whether the mechanism has anything to protect. The genotype does not change—only the signal it is fed— so the salience gate inherits the quality of its input, and choosing that input is a design decision as consequential as any axis of the space.

VIII. LIMITATIONS AND FUTURE WORK

The real-data validation of Section V uses a single extractor (ResNet-18 pretrained on ImageNet, PCA to 32 dimensions), a single dataset and a single capacity. We do not know whether the intra-class similarity of 0.401 is a property of the visual domain, of the extractor, or of both; a fine-tuned extractor, or a merge threshold recalibrated per domain, could give a

TABLE IV
MINIGRID-MEMORYS13-V0: THREE CONDITIONS, 40 ROLLOUTS EACH.

Policy	Surprise	Rare	Overlap	Frontier	FIFO
Random	Perception	38	75.4 %	0.0 %	2.6 %
Forward-biased	Perception	156	73.5 %	0.0 %	1.3 %
Random	Reward	38	5.6 %	52.6 %	2.6 %

different result. Silhouette-based prototype estimation, tight on synthetic streams, becomes uninformative here. The MiniGrid validation likewise uses one environment and one exploration policy besides the forward-biased one.

The most consequential open question is a tension between our strongest result and the setting that motivates the work. Section VI resolves the salience precondition by anchoring surprise in reward—but LOLA is defined by the *absence* of predefined goals, and an open-ended robot may face long stretches with no reward to predict. The available signal then falls back to perceptual novelty, which is precisely where we measured retention at the floor. The fix is valid where a reward exists and unproven where it does not, which is the regime e-MDB is ultimately built for. Intrinsic motivation, learning progress, or e-MDB’s own goal-babbling objectives are testable substitutes within the same protocol, since the strength gate is indifferent to where its scalar comes from; none is evaluated here. Finally, no result in this work was obtained on a physical robot: real sensor noise, continuous control and lifelong non-stationarity remain untested, and the generator’s distributional-drift parameter is never activated.

IX. CONCLUSION

We present EMBER, a research program on bioinspired episodic memory for cognitive robots, with four empirical contributions. Exhaustive architecture search shows that the relative importance of bioinspired memory mechanisms is not fixed but depends on the operating regime, characterized by the ratio between recurring experience prototypes and memory capacity. The result is two regimes: below a threshold near 1 the write axis governs the outcome, and above it the eviction policy does. Critically, the salience mechanism at encoding—the one the literature most motivates—shows a negligible main effect, yet conditioned on the single eviction policy that reads the strength signal it recovers fourteen times the rare-event retention. The effect is not weak; it is gated. The architecture benchmark sharpens this: SDM with a prediction-error gate is the only architecture combining high reconstruction fidelity with perfect rare-event retention, while the e-MDB FIFO reconstructs as well as any admitted architecture and is the floor at retention—its deficiency is not remembering, but choosing what to keep. The pure spiking circuit, validated in its native domain, does not transfer to continuous vectors, identifying PoC-5 as the next implementation milestone. Instrumenting the law on real embeddings exposes a precondition no synthetic design reveals: without sufficient intra-prototype similarity there is no ratio, nominal or effective, that produces a compression regime. Far from weakening the law, this turns it into an online-measurable quantity a

robot can use to decide whether consolidation is worthwhile. Finally, validating salience under partial observability reveals an analogous precondition and, unlike the first, resolves it: retention stays at the floor with perceptual surprise and rises to more than half when the signal is anchored in reward, without changing genotype or policy. The salience mechanism works in a real domain; what it needs is the right signal.

CODE AND DATA AVAILABILITY

The design space, evaluation tasks, six experiments and measured results (`results/**/*`.json, versioned alongside the code) backing every number in this paper will be released under an open license after review.

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